

Seeing the Invisible

Physics-Guided Missing-Physics Learning for PEMFC Flooding Prediction

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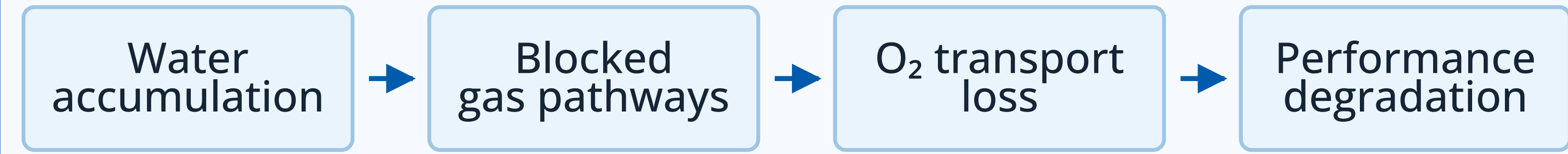


ABSTRACT

Liquid water accumulation in porous gas diffusion layers limits mass transport in polymer electrolyte membrane fuel cells, while uncertain constitutive relations remain a major source of prediction error. We propose a physics-informed grey-box framework that retains the governing equations in a finite-element solver and learns only the missing gas-phase transport closure. The closure is identified from synthetic reference responses using discrete-adjoint sensitivities without prescribing its functional form. After training at RH 0%, the learned closure is frozen and evaluated zero-shot at unseen RH 70% on 16×40 and 32×80 meshes. The recovered law reaches 2.16% relative L^2 error, while saturation-field errors remain below 0.56%. Field-level agreement is retained without retraining, supporting transfer across the tested humidity and discretization changes and establishing a foundation for physics-guided digital twins of PEMFC flooding.

INTRODUCTION

Why flooding matters



What limits flooding prediction?

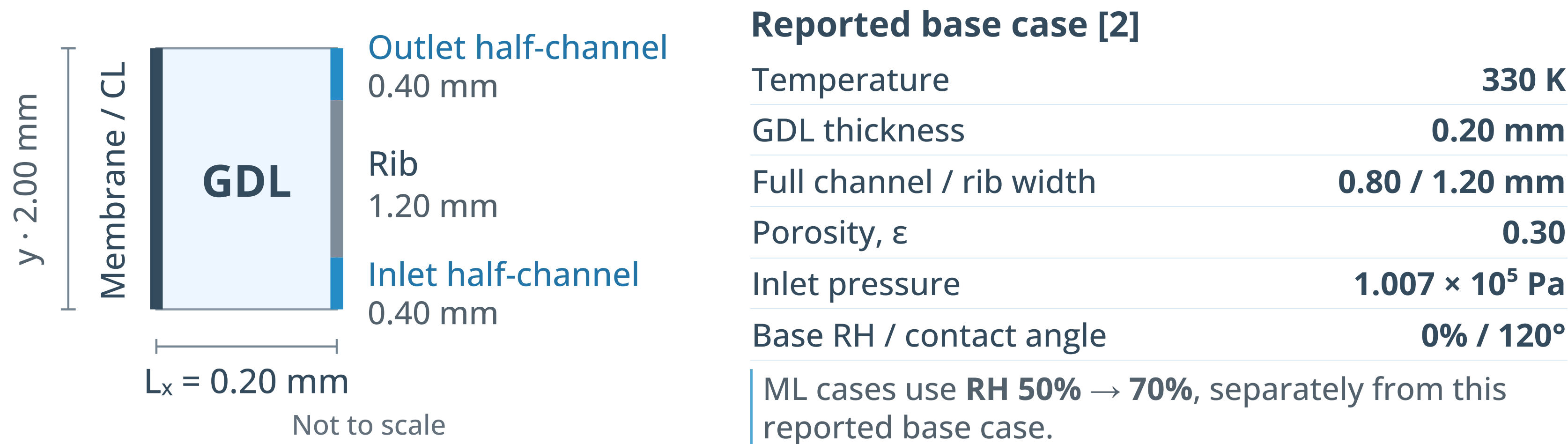
Observation	Internal liquid-water saturation $s(x,y)$ cannot be directly and continuously observed during operation.
Modeling	Flooding predictions depend on uncertain constitutive relations such as $D_{\text{eff}}(s)$, $k_r(s)$, and $p_c(s)$.

Existing ML vs our learning objective

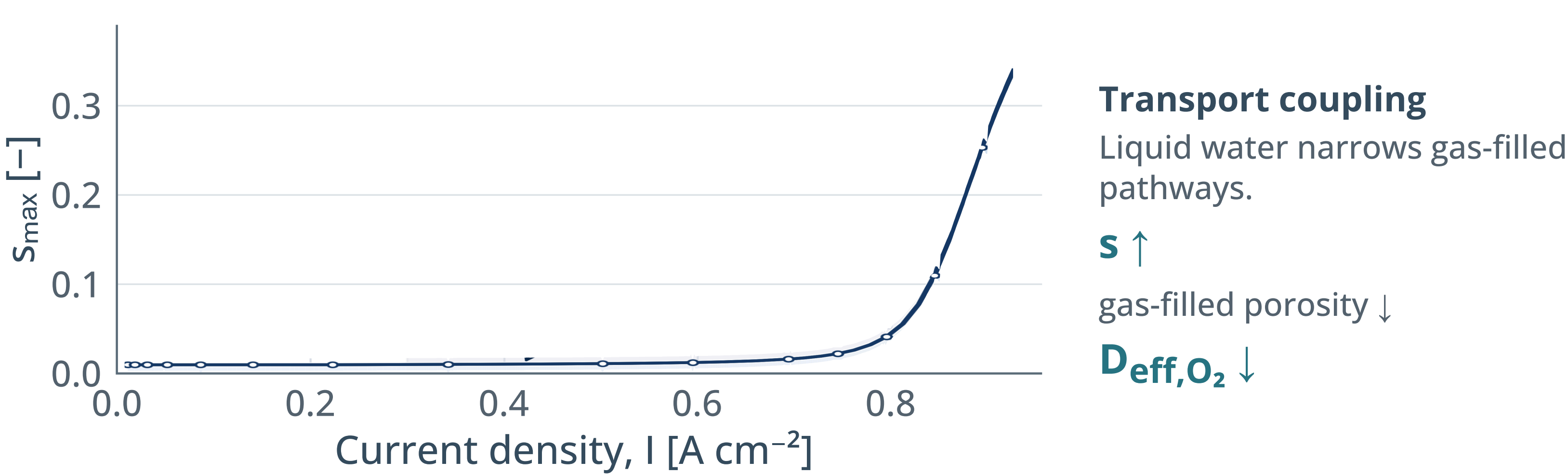
PINN Learn the PDE solution Physics is enforced through equation-residual training.	FNO / Neural operator Learn the solution operator Map input fields or parameters to solution fields.	OUR APPROACH FEM + Missing Physics ML Learn the constitutive law Only the uncertain transport relation is learned.
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RESULTS

01 Model domain & reference conditions



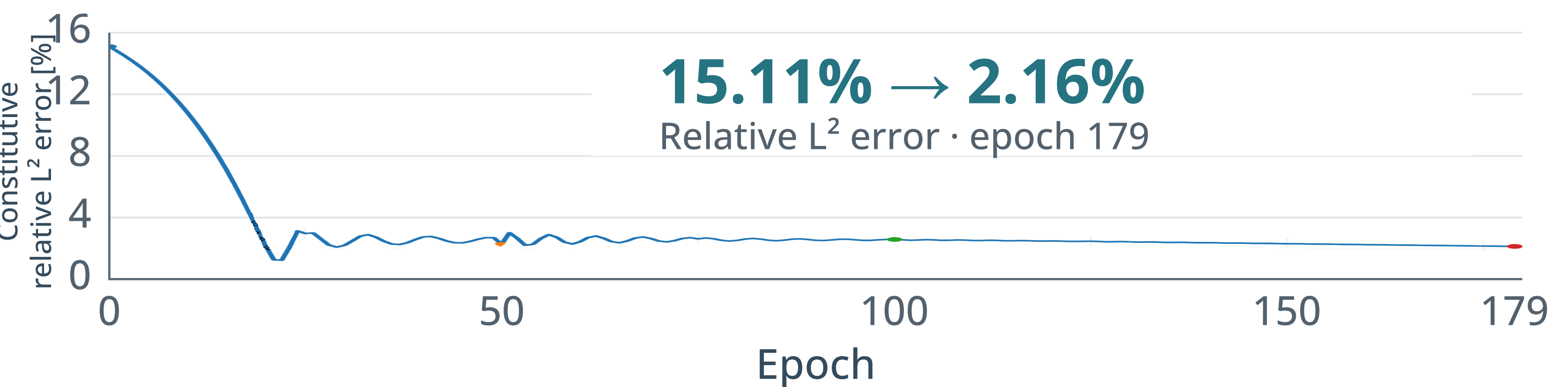
02 Water accumulation grows at high current



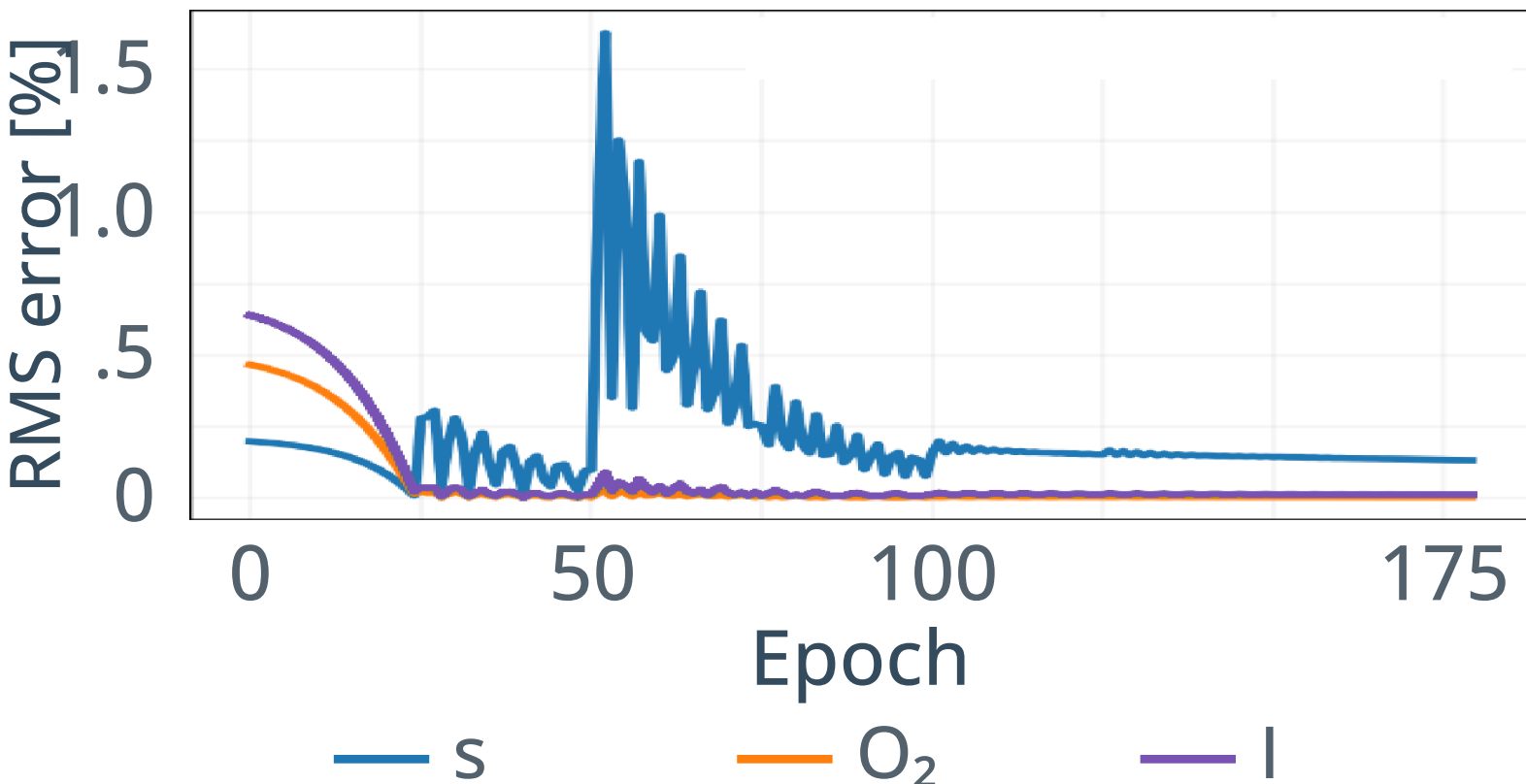
S_{max} : maximum liquid saturation within the GDL.

03 Recover the missing transport relation

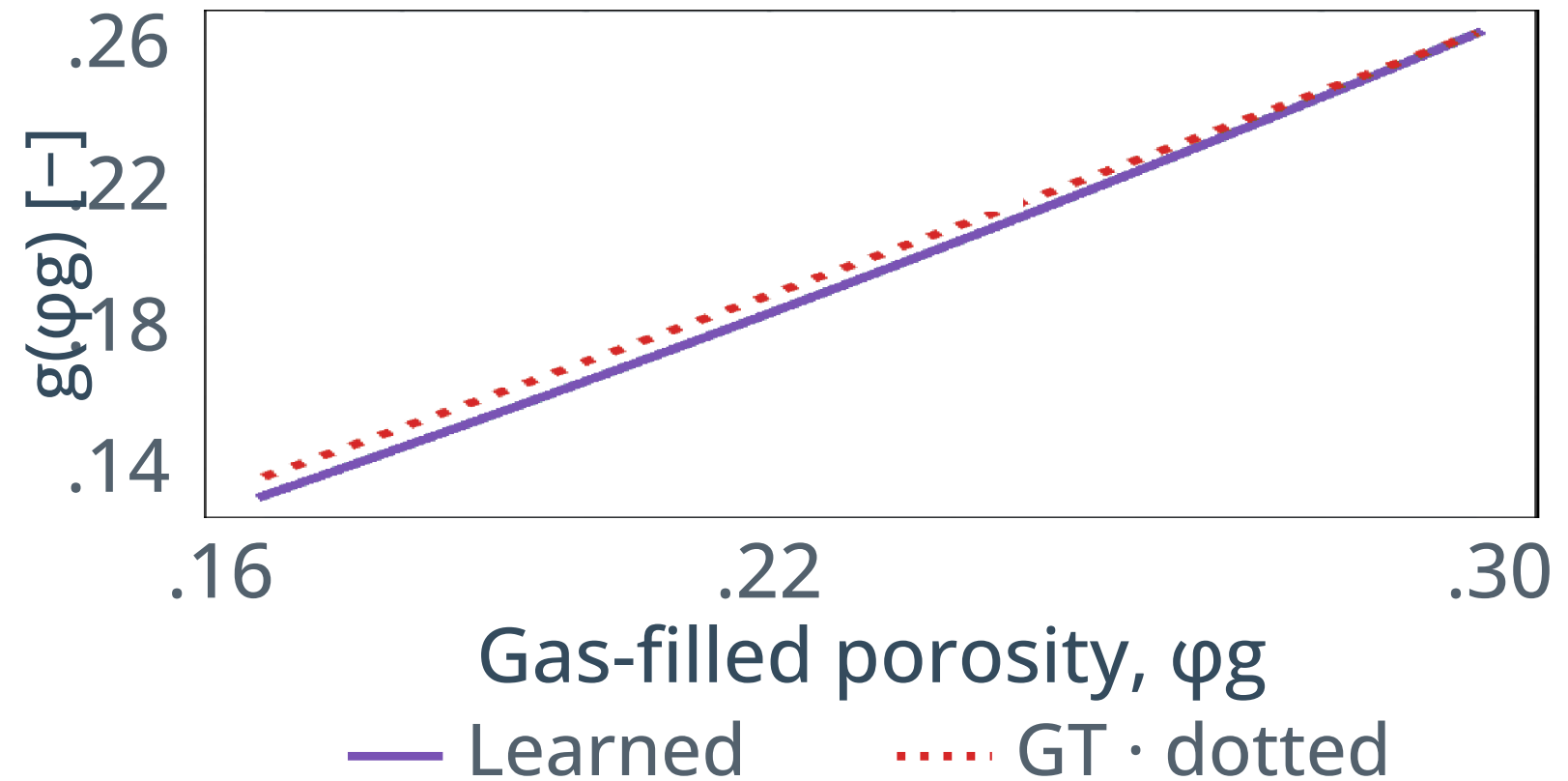
Train at RH 0%: infer the gas-phase closure from synthetic FEM responses.



State/response reconstruction

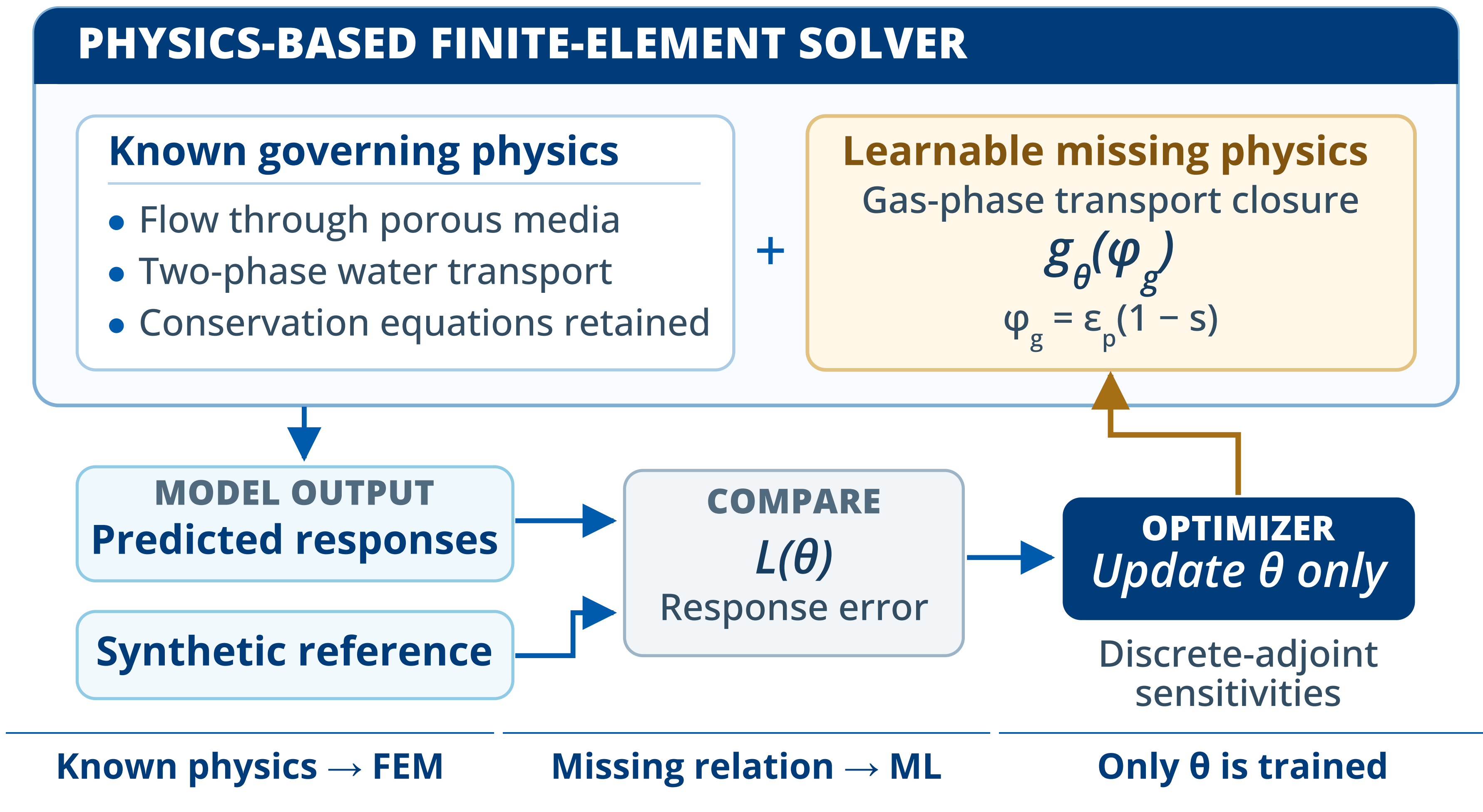


Recovered constitutive law

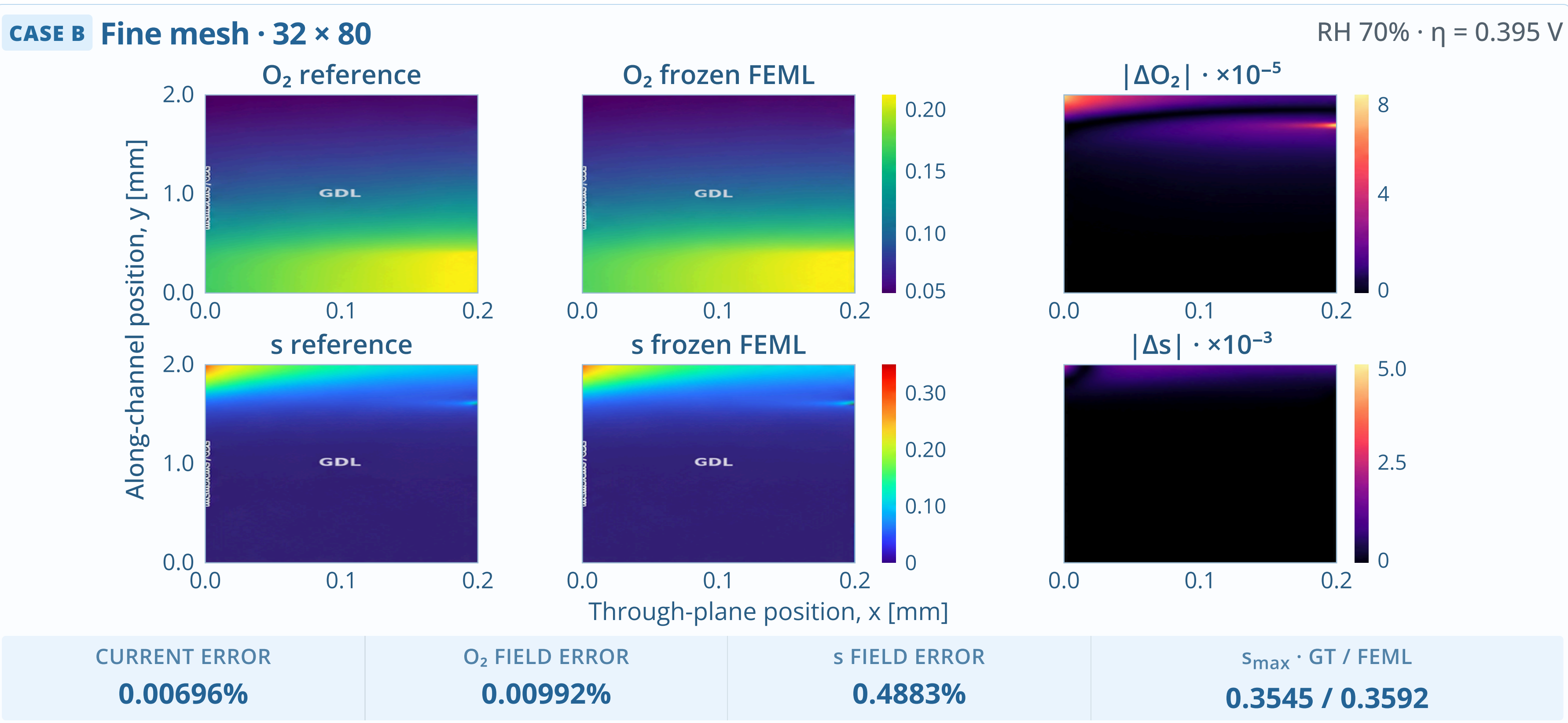
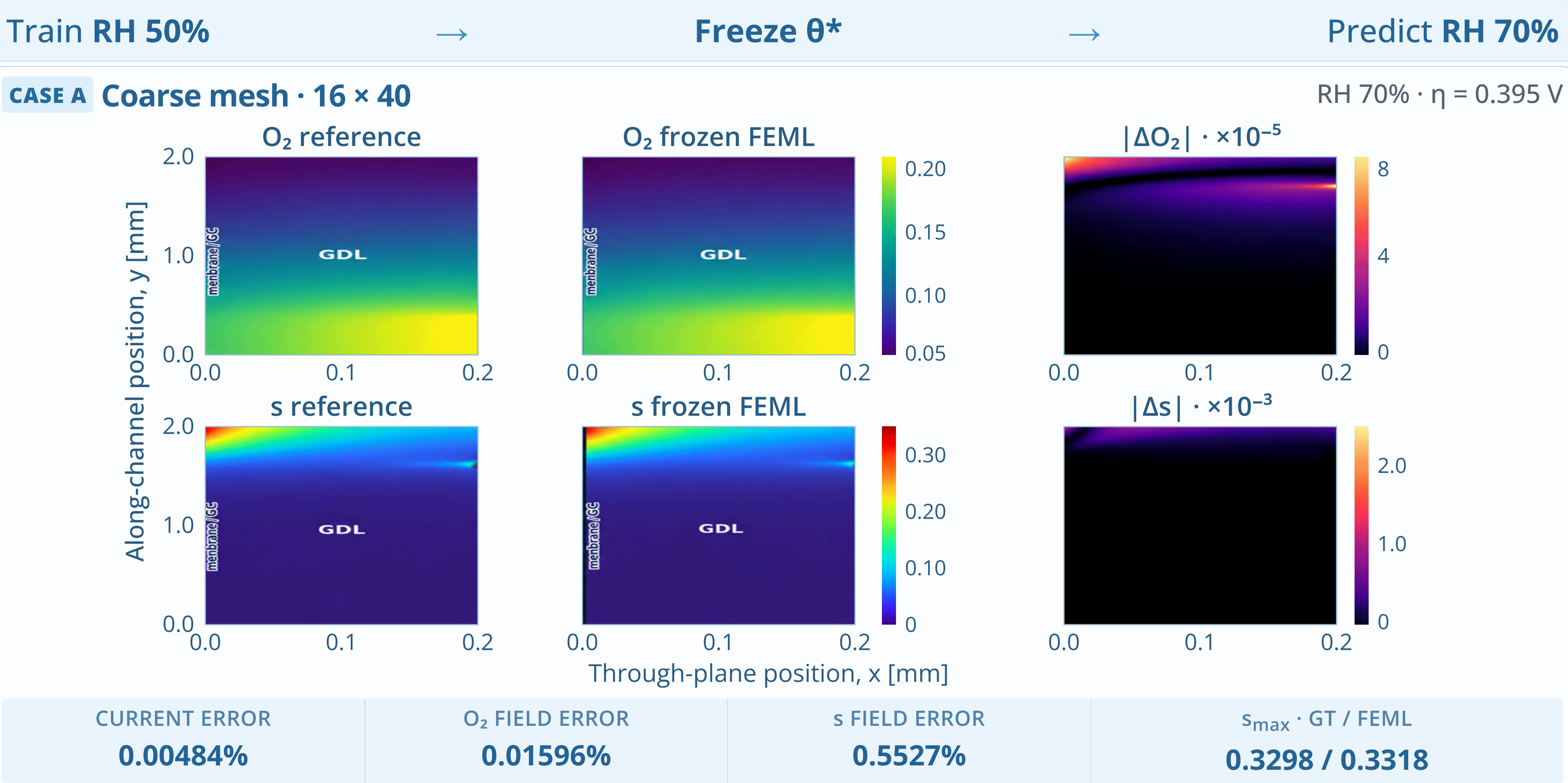


$D_{\text{eff}} = D_{\text{dry}} g(\phi_g)$ $\phi_g = \epsilon_p(1 - s)$ GT: $g(\phi_g) = \phi_g^\tau$

MODELING FRAMEWORK



04 Zero-shot transfer with a frozen closure



CONCLUSIONS & OUTLOOK

A learned transport law can be reused inside FEM without retraining.

01 Physically interpretable learning. The recovered closure links gas-filled porosity to transport, making the learned relation directly inspectable.

02 Transfer within the tested conditions. Agreement at changed RH and discretization supports reuse of the closure; it does not establish mesh convergence.

03 Next step: predictive validation. Test against experimental responses and a wider operating range, then assess closure identifiability and uncertainty.

REFERENCES [1] A. Farsi, N. Bouziani, D. A. Ham. Missing physics discovery through fully differentiable finite element-based machine learning. Scientific Reports (2026).
[2] M. Abdollahzadeh et al. Analysis of PEM fuel cell cathode two-dimensional modeling. Energy 68, 478–494 (2014).